

Numerical Results

31 March 2026

1 Experimental Setting

$$S_0 = 100, \quad v_0 = 0.114, \quad v'_0 = 0.110, \quad r = 0, \quad T = 1,$$

with

$$\kappa_1 = 5.5, \quad \kappa_2 = 0.1, \quad \theta = 0.078, \quad \xi_1 = 2.689, \quad \xi_2 = 0.502,$$

and

$$\delta_1 = \delta_2 = 0.94, \quad \rho_{12} = -0.982, \quad \rho_{13} = -0.727, \quad \rho_{23} = 0.59.$$

The Bermudan contract is sampled at 12 exercise dates. Five strike levels are considered:

$$K \in \{70, 80, 90, 100, 110\},$$

corresponding to deeper OTM through ITM puts.

All comparisons in this note use the fixed benchmark references generated with 1200 Euler steps. The reference direct prices are

$$V_{110}^{\text{ref}} = 16.645784, \quad V_{100}^{\text{ref}} = 11.356823, \quad V_{90}^{\text{ref}} = 7.402999,$$

$$V_{80}^{\text{ref}} = 4.596899, \quad V_{70}^{\text{ref}} = 2.699339.$$

For any direct price estimate V and reference value V^{ref} , the direct relative error is defined by

$$\varepsilon_{\text{dir}} = \frac{|V - V^{\text{ref}}|}{|V^{\text{ref}}|}.$$

The 95% confidence interval is reported as

$$V \pm 1.96 \text{ SE},$$

and the confidence-interval width is therefore $2 \times 1.96 \times \text{SE}$.

The objective is to compare plain LSMC with the Hybrid LSMC–PDE method under *matched path budgets*. Two matched-path regimes are examined:

- 60,000 direct paths for both methods;
- 20,000 direct paths for both methods.

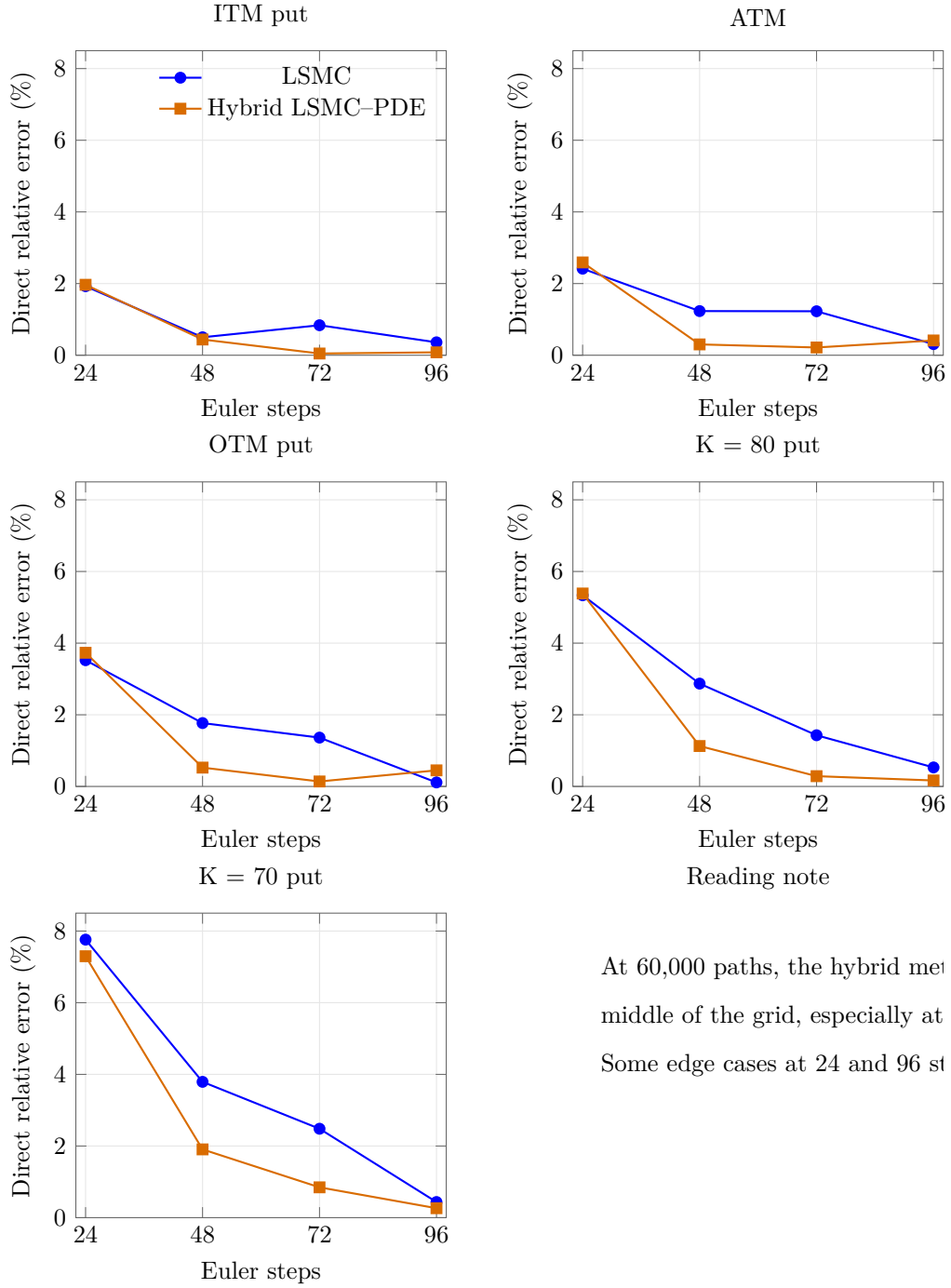
The Euler step counts are 24, 48, 72, 96. The hybrid numerical setting is kept fixed at 301 asset grid points, asset range factors 0.30 and 3.50, and volatility truncation quantile 0.999.

2 Matched 60,000-Path Comparison

Table 1 shows the Euler steps at which the Hybrid LSMC–PDE method achieves a smaller direct error than plain LSMC when both methods use 60,000 direct paths.

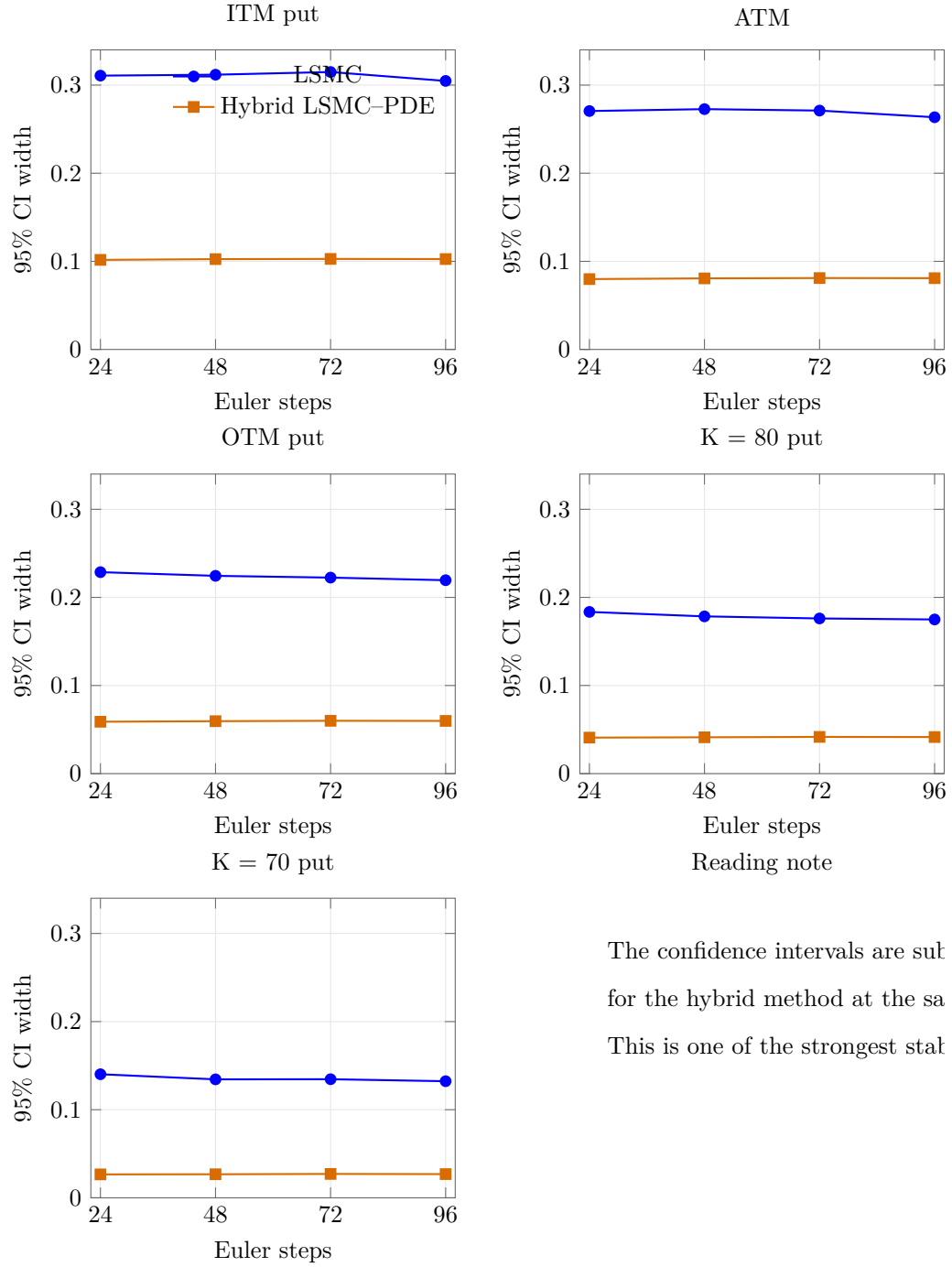
The main favorable region is the middle of the step grid, especially 48 to 72 steps.

Table 2 reports representative results at 72 steps, including direct prices, 95% confidence intervals, and runtime.



At 60,000 paths, the hybrid method shows lower error than the LSMC method, especially at the middle of the grid, especially at 48 and 72 steps. Some edge cases at 24 and 96 steps.

Figure 1: Direct relative error under the matched 60,000-path comparison.



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Figure 2: 95% confidence-interval width under the matched 60,000-path comparison.

Table 1: Hybrid win counts under the matched 60,000-path comparison.

Scenario	Hybrid better at steps
ITM put	48, 72, 96
ATM	48, 72
OTM put	48, 72
K = 80 put	48, 72, 96
K = 70 put	24, 48, 72, 96

3 Matched 20,000-Path Comparison

The matched path budget was then reduced from 60,000 to 20,000 for both methods.

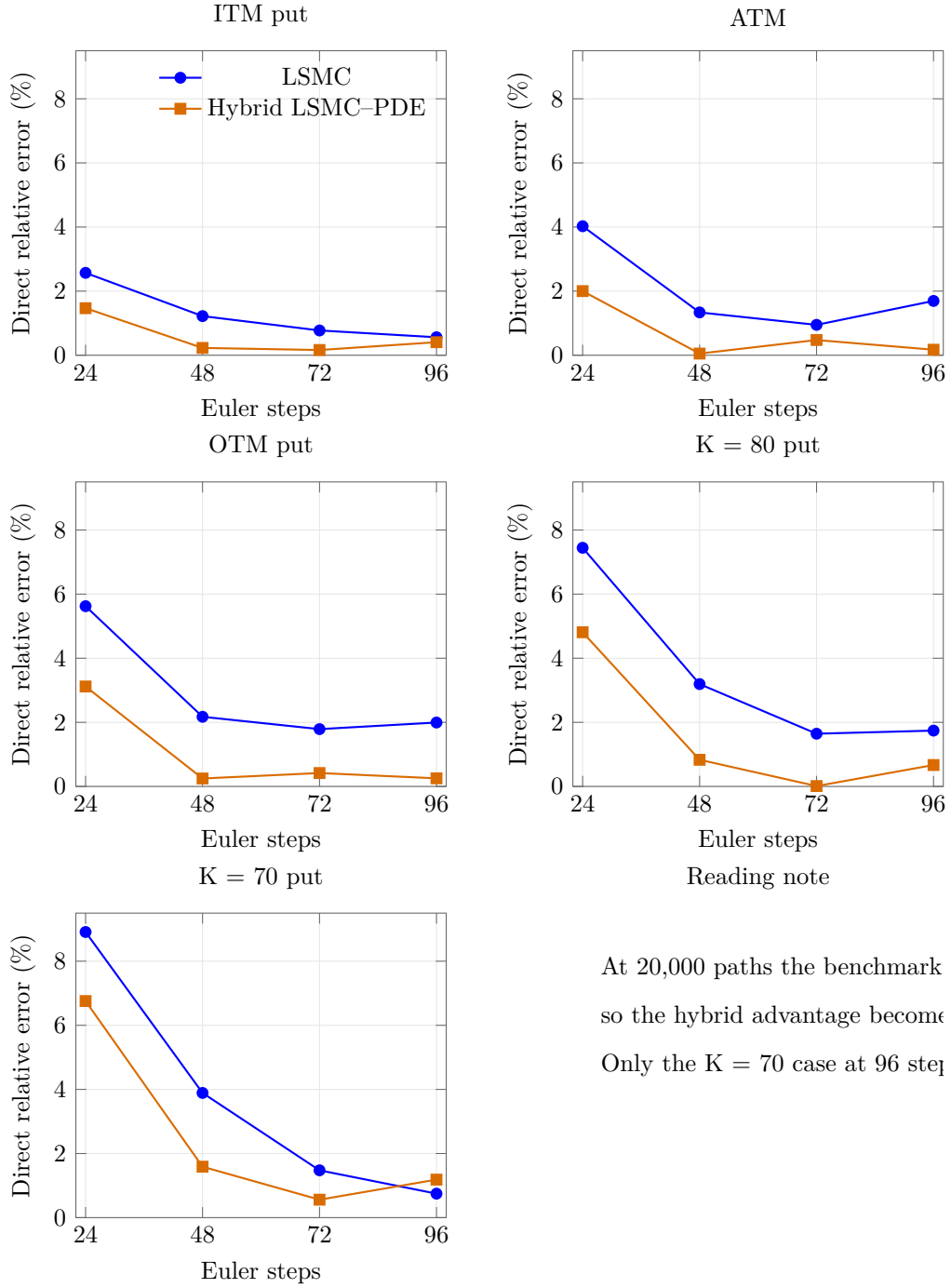
Reducing the path budget strengthens the favorable direct-error result. The benchmark LSMC method deteriorates faster than the hybrid method, so the hybrid advantage becomes broader across the step grid.

Table 4 reports representative 72-step values for the 20,000-path comparison.

4 Conclusions

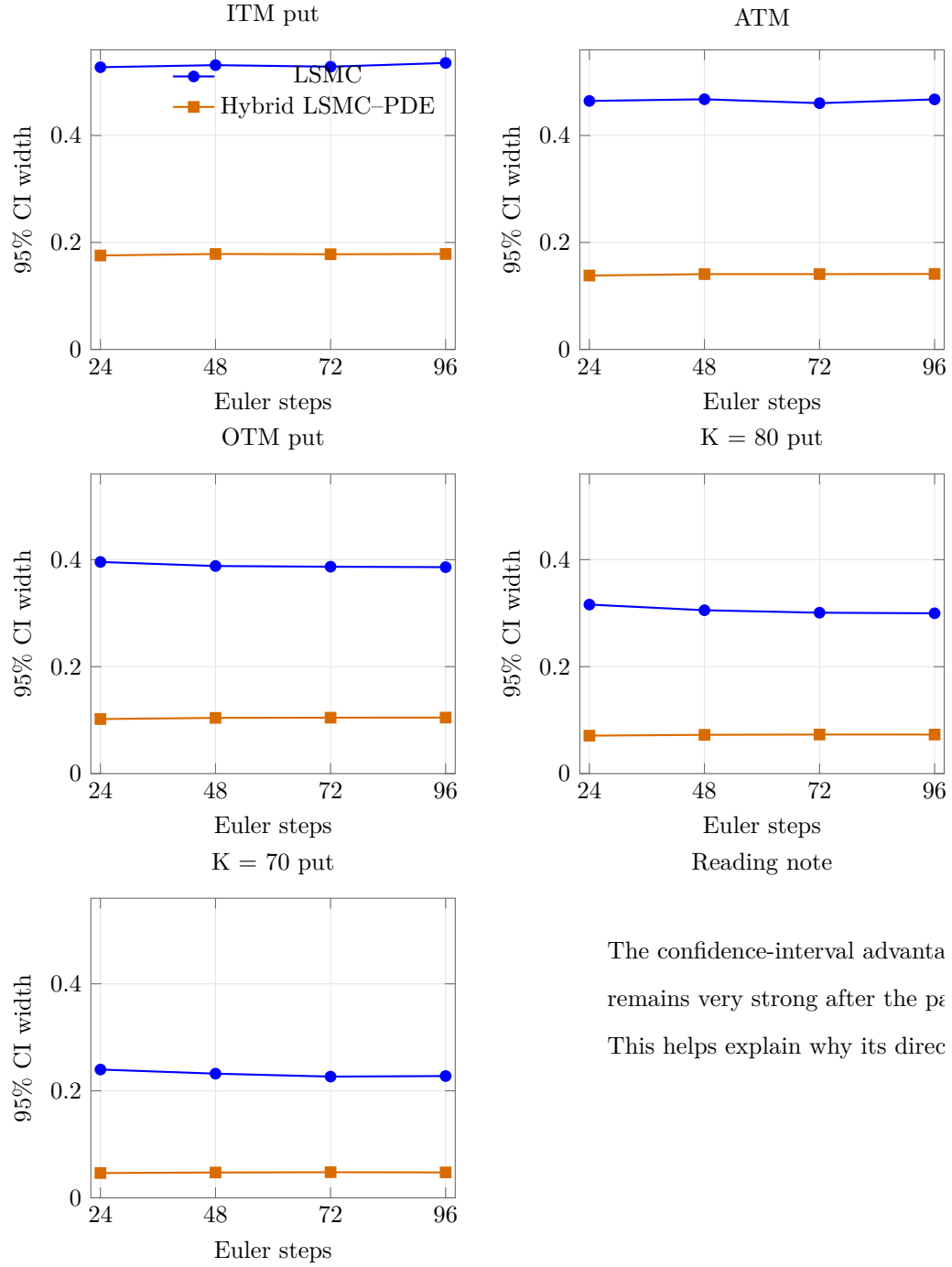
- Under the matched 60,000-path comparison, the Hybrid LSMC–PDE method is most favorable in the middle of the Euler grid, especially at 48 and 72 steps. Some edge cases at 24 and 96 steps still favor plain LSMC.
- Under the matched 20,000-path comparison, the favorable hybrid result becomes stronger. The benchmark LSMC method deteriorates faster, and the hybrid method becomes more accurate in almost all tested cases.
- In both matched-path regimes, the hybrid method has much smaller standard errors and much narrower confidence intervals across the full strike range. This is one of the clearest robust findings of the study.
- The main cost of the hybrid method remains runtime. Plain LSMC is substantially faster in both cases, even when it is less accurate and less stable at equal path budgets.

Taken together, the two matched-path studies show a consistent pattern. When the comparison is made on a fair equal-path basis, the Hybrid LSMC–PDE method performs particularly well at moderate Euler resolutions. Moreover, when the path budget is tightened, the hybrid advantage becomes even more visible because its confidence intervals remain much narrower than those of plain LSMC.



At 20,000 paths the benchmark
so the hybrid advantage become
Only the K = 70 case at 96 step

Figure 3: Direct relative error under the matched 20,000-path comparison.



The confidence-interval advantage
remains very strong after the price
This helps explain why its direction

Figure 4: 95% confidence-interval width under the matched 20,000-path comparison.

Table 2: Representative 72-step values under the matched 60,000-path comparison.

Scenario	LSMC			Hybrid LSMC–PDE		
	Direct price	95% CI	Runtime	Direct price	95% CI	Runtime
ITM put	16.785116	[16.627687, 16.942545]	1.18 s	16.653809	[16.602442, 16.705177]	238.7
ATM	11.496193	[11.360645, 11.631741]	1.25 s	11.332296	[11.291787, 11.372805]	241.8
OTM put	7.503911	[7.392681, 7.615141]	1.16 s	7.392732	[7.362757, 7.422707]	240.1
K = 80 put	4.662578	[4.574527, 4.750629]	1.17 s	4.610083	[4.589249, 4.630917]	230.5
K = 70 put	2.766338	[2.699045, 2.833631]	1.21 s	2.722197	[2.708657, 2.735737]	233.9

Table 3: Hybrid win counts under the matched 20,000-path comparison.

Scenario	Hybrid better at steps
ITM put	24, 48, 72, 96
ATM	24, 48, 72, 96
OTM put	24, 48, 72, 96
K = 80 put	24, 48, 72, 96
K = 70 put	24, 48, 72

Table 4: Representative 72-step values under the matched 20,000-path comparison.

Scenario	LSMC			Hybrid LSMC–PDE		
	Direct price	95% CI	Runtime	Direct price	95% CI	Runtime
ITM put	16.774434	[16.510179, 17.038689]	0.43 s	16.618575	[16.529605, 16.707546]	76.2
ATM	11.464532	[11.234369, 11.694695]	0.46 s	11.302836	[11.232454, 11.373219]	78.0
OTM put	7.535388	[7.342003, 7.728773]	0.50 s	7.372034	[7.319752, 7.424315]	78.7
K = 80 put	4.672537	[4.522131, 4.822943]	0.49 s	4.597431	[4.560886, 4.633975]	78.6
K = 70 put	2.739173	[2.625881, 2.852465]	0.50 s	2.714444	[2.690521, 2.738368]	78.6